26/05/2026

CODE-P

Aakash
Medical | IIT-JEE | Foundations

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# PROJECT PHOENIX FOR RE-NEET-2026

**MM: 720** **CST – 01P** **Time: 180 Mins.**

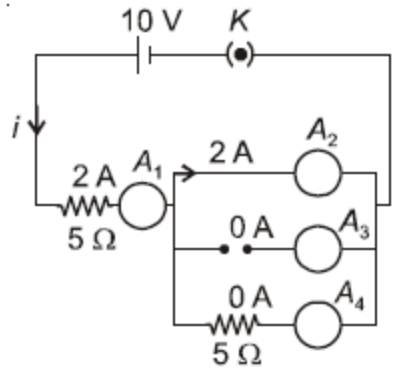
## Answers

| | | | | |
|---|---|---|---|---|
| 1. (3) | 37. (1) | 73. (3) | 109. (3) | 145. (2) |
| 2. (1) | 38. (1) | 74. (3) | 110. (4) | 146. (1) |
| 3. (4) | 39. (2) | 75. (4) | 111. (2) | 147. (2) |
| 4. (4) | 40. (1) | 76. (4) | 112. (4) | 148. (2) |
| 5. (1) | 41. (1) | 77. (1) | 113. (1) | 149. (3) |
| 6. (2) | 42. (4) | 78. (4) | 114. (2) | 150. (4) |
| 7. (1) | 43. (1) | 79. (2) | 115. (3) | 151. (2) |
| 8. (2) | 44. (3) | 80. (3) | 116. (2) | 152. (4) |
| 9. (2) | 45. (4) | 81. (1) | 117. (2) | 153. (3) |
| 10. (1) | 46. (4) | 82. (3) | 118. (2) | 154. (3) |
| 11. (3) | 47. (2) | 83. (3) | 119. (3) | 155. (1) |
| 12. (3) | 48. (3) | 84. (4) | 120. (4) | 156. (4) |
| 13. (3) | 49. (1) | 85. (3) | 121. (2) | 157. (1) |
| 14. (4) | 50. (2) | 86. (4) | 122. (2) | 158. (3) |
| 15. (1) | 51. (3) | 87. (3) | 123. (4) | 159. (4) |
| 16. (4) | 52. (3) | 88. (4) | 124. (3) | 160. (3) |
| 17. (1) | 53. (1) | 89. (3) | 125. (4) | 161. (1) |
| 18. (3) | 54. (2) | 90. (4) | 126. (2) | 162. (1) |
| 19. (1) | 55. (4) | 91. (1) | 127. (4) | 163. (2) |
| 20. (1) | 56. (2) | 92. (3) | 128. (2) | 164. (2) |
| 21. (3) | 57. (2) | 93. (4) | 129. (2) | 165. (1) |
| 22. (4) | 58. (4) | 94. (1) | 130. (1) | 166. (3) |
| 23. (3) | 59. (2) | 95. (2) | 131. (2) | 167. (4) |
| 24. (2) | 60. (3) | 96. (4) | 132. (3) | 168. (4) |
| 25. (3) | 61. (2) | 97. (2) | 133. (3) | 169. (3) |
| 26. (4) | 62. (3) | 98. (2) | 134. (2) | 170. (4) |
| 27. (4) | 63. (3) | 99. (2) | 135. (4) | 171. (1) |
| 28. (1) | 64. (1) | 100. (2) | 136. (4) | 172. (1) |
| 29. (1) | 65. (3) | 101. (1) | 137. (3) | 173. (3) |
| 30. (1) | 66. (3) | 102. (4) | 138. (1) | 174. (1) |
| 31. (2) | 67. (2) | 103. (1) | 139. (1) | 175. (3) |
| 32. (2) | 68. (4) | 104. (2) | 140. (1) | 176. (4) |
| 33. (4) | 69. (4) | 105. (3) | 141. (1) | 177. (2) |
| 34. (3) | 70. (4) | 106. (3) | 142. (2) | 178. (4) |
| 35. (1) | 71. (4) | 107. (1) | 143. (4) | 179. (2) |
| 36. (3) | 72. (1) | 108. (4) | 144. (2) | 180. (3) |

## Hints and Solutions

### PHYSICS

**(1)** **Answer :** (3)
**Hint:**
Inductor behaves like short circuit and capacitor behaves like open circuit at $t = \infty$
**Solution:**
After long time, circuit will be like




$i = \frac{10}{5} = 2\ A$
Hence reading of $A_1$ and $A_2$ = 2 A
Reading of $A_3$ and $A_4$ = 0
Hence $A_1$, $A_2$, $A_3$ and $A_4$ → 2 A, 2 A, 0, 0

**(2)** **Answer :** (1)
**Solution:**
Internal force cannot change the state of motion of the system.

**(3)** **Answer :** (4)
**Solution:**
$P_{out}$ = 50% of $P_{in}$
$2000 \times i = \frac{50}{100} \times 10 \times 100$
$i = \frac{50}{100} \times \frac{10 \times 100}{2000} = 0.25\ A$

**(4)** **Answer :** (4)
**Solution:**
$r \propto n^2$
$\frac{R}{R_1} = \frac{4}{16} \Rightarrow R_1 = 4R$

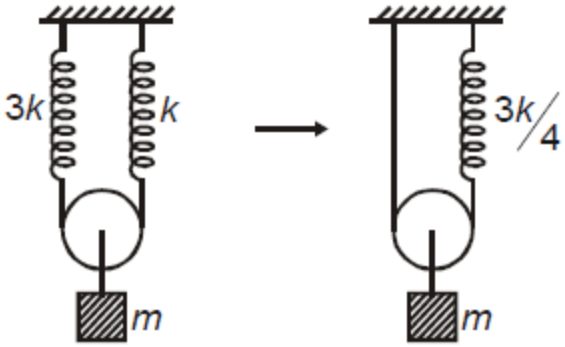
**(5)** **Answer :** (1)
**Solution:**
$\frac{1}{2}mv^2 - \frac{GMm}{R} = \frac{-GMm}{R+h}$
$v = \sqrt{\frac{Gm}{R}}$
$\frac{1}{2}\ \frac{GMm}{R} - \frac{GMm}{R} = \frac{-GMm}{R+h}$
$\frac{1}{2R} = \frac{1}{R+h}$
$R = h$

**(6)** **Answer :** (2)
**Solution:**
Impedance Z = $\sqrt{(R^2 + X^2)}$.
use PF = R/Z

**(7)** **Answer :** (1)
**Solution:**

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Answer (1)



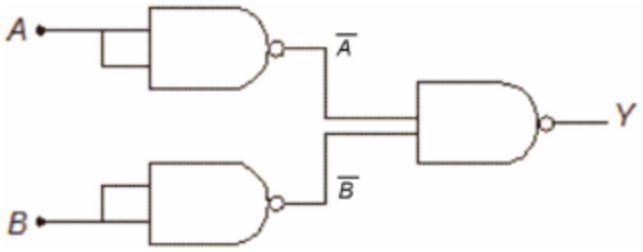

$$T = 2\pi\sqrt{\frac{m}{4k_{eq}}} = 2\pi\sqrt{\frac{m\times 4}{4\times 3k}} = 2\pi\sqrt{\frac{m}{3k}}$$

**(8)** **Answer :** (2)
**Solution:**

$\frac{I_1}{I_2} = \frac{25}{1}$

$\frac{I_{max}}{I_{min}} = \frac{(\sqrt{I_1}+\sqrt{I_2})^2}{(\sqrt{I_1}-\sqrt{I_2})^2} = \frac{36}{16} = \frac{9}{4}$

**(9)** **Answer :** (2)
**Solution:**

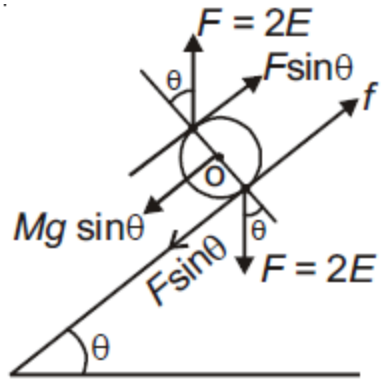



$Y = \overline{\overline{A}\,\overline{B}} = A + B$

**(10)** **Answer :** (1)
**Solution:**

$P = \frac{n}{V}RT$ when *n* and *V* are same $P \propto T$

**(11)** **Answer :** (3)
**Solution:**




*O* is center of ring
$F_{net} = 0$
$Mg \sin\theta = f$ ....(i)

$\left[\tau_{Net}\right]_0 = 0$

$2R\,F\sin\theta - R\,f = 0$
$2F\sin\theta = f$ ....(ii)

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From (i) and (ii)

$F = \frac{Mg}{2}$

$\Rightarrow 2E = \frac{Mg}{2}$

⇒ $E$ = 5 V/m

**(12)** **Answer :** (3)
**Solution:**
**Hint & Sol.:** $v = \frac{c}{\mu}$

**(13)** **Answer :** (3)
**Hint:**
Power ($P$) transmitted in the beam = $NE$ = $Nhf$
**Solution:**
$P = 2 \times 10^{16} \times 6.63 \times 10^{-34} \times 5 \times 10^{14}$
$= 6.63 \times 10^{-3}$ W = 6.63 mW

**(14)** **Answer :** (4)
**Solution:**
$I_2 = \frac{5}{1} = 5\ \text{mA}$
$I = \frac{22-5}{2} = 8.5\ \text{mA}$
$I_Z = I - I_2$
= 8.5 – 5 = 3.5 mA

**(15)** **Answer :** (1)
**Hint:**
Diode $D_1$ is reversed biased and diode $D_2$ is forward biased.
**Solution:**
Current will flow through diode $D_2$ only i.e. through 40 Ω resistor.
current = $\frac{4}{40} = 0.1\ \text{A}$

**(16)** **Answer :** (4)
**Solution:**
Net force at equilibrium is zero

**(17)** **Answer :** (1)
**Hint:**
$\lambda_{micro} > \lambda_{IR} > \lambda_{uv} > \lambda_{gamma}$
**Solution:**
$\lambda_{micro} > \lambda_{IR} > \lambda_{uv} > \lambda_{gamma}$

**(18)** **Answer :** (3)
**Solution:**
Nuclear density of all nuclei are nearly constant.

**(19)** **Answer :** (1)
**Solution:**

$\because \quad [U_E]_{av.} = \frac{\varepsilon_0 E_0^2}{4} = \frac{1}{4} \times 8.85 \times 10^{-12} \times (2)^2$

$[U_E]_{av.} = 8.85 \times 10^{-12} \text{J}/m^3$

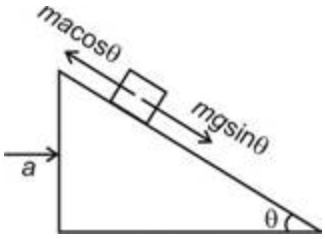
$\because \quad [U_B]_{av.} = U_E = 8.85 \times 10^{-12}$

$\therefore \quad U_{av.} = [U_E]_{av.} + [U_B]_{av.}$

$V_{av.} = 17.70 \times 10^{-12}$ J/m$^3$

**(20)** **Answer :** (1)
**Solution:**

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$\sin\theta = \frac{1}{\sqrt{2}} \therefore \theta = 45^\circ$

In equilibrium
*mg*sinθ = *ma*cosθ (no relative slip)
*a* = *g*tanθ
*a* = *g*

**(21) Answer :** (3)
**Solution:**

$F = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k}$

$= -\left[\frac{\partial}{\partial x}\frac{10xz}{y}\hat{i} + \frac{\partial}{\partial y}\frac{10xz}{y}\hat{j} + \frac{\partial}{\partial z}\frac{10xz}{y}\hat{k}\right]$

$= -\frac{10z}{y}\hat{i} + \frac{10xz}{y^2}\hat{j} - \frac{10x}{y}\hat{k}$

**(22) Answer :** (4)
**Solution:**
The electric potential due to a point charge is given by V = k * q / r, where k is Coulomb's constant, q is the charge, and r is the distance from the charge to the point of interest.

**(23) Answer :** (3)
**Solution:**

Using the lens' formula, $\frac{1}{f_{air}} = (\mu - 1)\left[\frac{1}{R} - \left(\frac{-1}{R}\right)\right]$

⇒ $R = 2f_{air}(\mu - 1) = 30(\mu - 1)$
(a-ii) μ = 1.4 ⇒ $R$ = 12 cm
(b-iii) μ = $\frac{3}{2}$ ⇒ $R$ = 15 cm
(c-i) μ = $\frac{4}{3}$ ⇒ $R$ = 10 cm
(d-iv) μ = 2.4 ⇒ $R$ = 42 cm

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**(24) Answer :** (2)
**Solution:**
$L = f_o + f_e$

$m = \frac{f_0}{f_{te}} = 19$

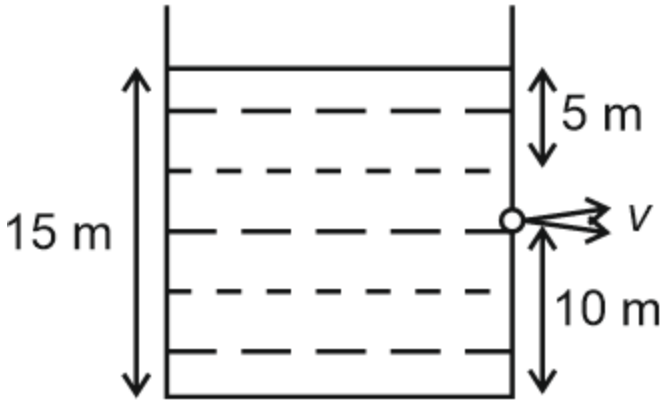
$19f_e = f_o$ …(1)
$100 = f_e + 19f_e$
⇒ $f_e = 5$
∴ $f_o$ = 95 cm

**(25) Answer :** (3)
**Solution:**

$v_{sound} \propto \sqrt{\text{Elasticity}}$

**(26) Answer :** (4)
**Solution:**
By Fleming's right hand rule.

**(27) Answer :** (4)
**Solution:**




$v = \sqrt{2gh} = \sqrt{2 \times 10 \times 5}$ ms$^{-1}$

= 10 ms$^{-1}$

**(28)** **Answer :** (1)
**Solution:**
$g_e < g_P$

$g_p = \frac{GM}{R_p{}^2}, g_\infty = \frac{GM}{R_\infty{}^2} (R_p < R_\infty)$

$Mg_e < Mg_P$ weight increases

**(29)** **Answer :** (1)
**Solution:**
Point source is placed at focus of convex lens, so after refraction light rays become parallel. Hence on changing the distance, there is no change in number of photons striking on photocell. So saturated current and cut-off potential remains same.

**(30)** **Answer :** (1)
**Solution:**
$m_1 : m_2 = 2 : 3$

$m = 20$ kg

Shell mass

$m_1 = 8$ kg, $m_2 = 12$ kg

From law of conservation of momentum

$0 = m_1v_1 + m_2v_2$

$0 = 8 \times 6 + 12 \times v$

$v_2 = \frac{48}{12} = 4$ m/s

KE larger fragment $= \frac{1}{2} \times 12 \times 16 = 12 \times 8 = 96$ J

**(31)** **Answer :** (2)
**Solution:**
$y_{cm} = \frac{R \sin \alpha}{\alpha}$ $\alpha = \frac{\pi}{2}$

$= \frac{2R}{\pi}$ if $R = 1$ cm

$y_{cm} = \frac{2}{\pi}$

**(32)** **Answer :** (2)
**Solution:**
$S_{last} = \frac{g}{2} = 5$ m

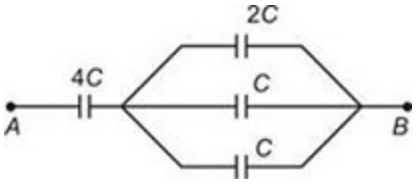
**(33)** **Answer :** (4)
**Hint:**
$\frac{dQ}{dt} = \sigma A(T^4 - T_0{}^4)$
**Solution:**
$\frac{dQ}{dt} = \sigma A(T^4 - T_0{}^4)$

$R = \sigma A(400^4 - 300^4)$

$R' = \sigma A(500^4 - 300^4)$

$\frac{R'}{R} = \frac{500^4 - 300^4}{400^4 - 300^4} = \frac{544}{175}$

**(34) Answer :** (3)
**Solution:**




$C_{AB} = \frac{(4C)(4C)}{8C} = 2C$

**(35) Answer :** (1)
**Solution:**
$R \propto \ell$

$\frac{R_1}{R_2} = \frac{\ell_1}{\ell_2} = \frac{1}{2}$ $\quad \therefore \quad R_1 = \frac{1}{3} \times 9 = 3\ \Omega$

$R_2 = \frac{2}{3} \times 9 = 6\ \Omega$

In parallel $R = \frac{R_1 R_2}{R_1 + R_2} = \frac{6 \times 3}{6+3} = \frac{18}{9} = 2\ \Omega$

**(36) Answer :** (3)
**Solution:**
As magnetic force is a result of cross product of velocity and magnetic field, it is perpendicular to both. When the angle between velocity and magnetic field is zero or 180°, then the magnetic force is zero.

**(37) Answer :** (1)
**Solution:**

$B_{\text{axial point}} = 2\left(\frac{\mu_0}{4\pi}\right)\frac{M}{r^3}$

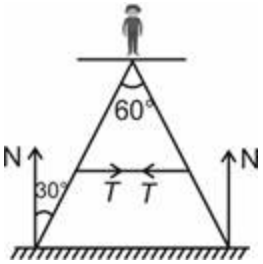
$= \frac{2 \times 10^{-7} \times 8}{(0.2)^3} = 2 \times 10^{-4}\text{T}$

$B_{\text{eq. point}} = \left(\frac{\mu_0}{4\pi}\right)\frac{M}{r^3} = 10^{-4}\text{T}$

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**(38) Answer :** (1)
**Solution:**
As $P \propto Fv$
$P' \propto 2F \times 2v$
$\Rightarrow \quad \boxed{P' = 4P}$

**(39) Answer :** (2)
**Solution:**




In equilibrium
Net vertical forces
2N = 40 × 10
N = 200 N
Take torque about upper end.
N × 2 tan 30° = $T$× 1
$T$ = 200 × 2 × $\frac{1}{\sqrt{3}}$
= 231 N

**(40) Answer :** (1)
**Solution:**

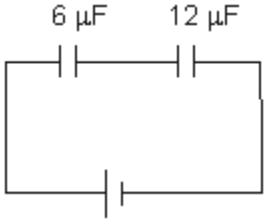
$\tau = I \propto$
$5 = (50) \propto$
$\propto \ = 0.1 \text{ rad}/\text{s}^2$

**(41) Answer :** (1)
**Solution:**
$R_{ex} = \frac{600\times1200}{600+1200} + 300 = 700$
$V = \frac{7\times400}{700} = 4 \text{ V}$

**(42) Answer :** (4)
**Solution:**
$\vec{r}_1 = 2\hat{i}, \vec{r}_2 = 3\hat{j}$
$\vec{r}_2 - \vec{r}_1 = 3\hat{j} - 2\hat{i}$
$|\vec{r}_2 - \vec{r}_1| = \sqrt{3^2+2^2} = \sqrt{13}$

**(43) Answer :** (1)
**Solution:**
$\frac{T_1}{T_2} = 3$
$\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{1}{3} = \frac{2}{3}$

**(44) Answer :** (3)
**Solution:**




Effective capacitance in series
$C$ = 4 μF
Maximum charged that can be stored by
6 μF = 6 × 200 = 1200 μC.
Maximum charged that can be stored by
12 μC = 12 × 200 = 2400 μC.
$q_6 < q_{12}$
So maximum charge given to series combination = 1200 μC.
$V_{\max} = \frac{q}{C} = \frac{1200}{4} = 300 \text{ V}$

**(45) Answer :** (4)
**Solution:**
**Hint:** breaking stress $\sigma = \frac{F}{A}$
**Sol.:** $\sigma_1 = \sigma_2$
$\Rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2}$
$\Rightarrow \frac{F}{A} = \frac{F'}{4A}$
$\Rightarrow F' = 4F$

## CHEMISTRY

**(46) Answer :** (4)
**Hint:**
Atomic mass of sulphur is 32 u
**Solution:**
For minimum molecular mass at least one sulphur atom should be present
Mass percentage of an element = $\frac{\text{Mass of an element}}{\text{Molecular mass}} \times 100$

$8 = \frac{32}{\text{Molecular mass}} \times 100$

Molecular mass = 400 u

**(47) Answer :** (2)
**Solution:**

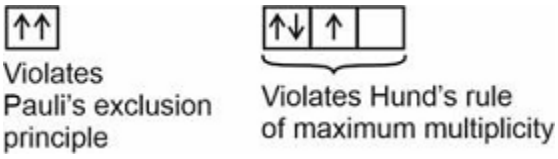
$\Delta E = 13.6 \times 1 \times \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$= 13.6 \times 1 \times \frac{8}{9} = 12.08\text{eV}$

$\phi$= 6.08 eV

$\therefore KE_{max} = 12.08 - 6.08 = 6$ eV

**(48) Answer :** (3)
**Hint:**
According to Pauli's exclusion principle, in an orbital no two electrons can have the same set of all four quantum numbers.
**Solution:**




**(49) Answer :** (1)
**Hint:**

Atomic number 36 is of Krypton (Kr) and 48 is of cadmium (Cd) and both are $5^{th}$ period elements.
**Solution:**

Z [49] $5s^2\ 5p^1$, $5^{th}$ period
group number = 10 + 2 + 1 = 13
Z [92] $5f^3\ 6d^1\ 7s^2$, $7^{th}$ period
All *f*-block elements are in $3^{rd}$ group
Z [38] $5s^2$, $5^{th}$ period and $2^{nd}$ group
Z [57] $5d^1\ 6s^2$, $6^{th}$ period and $3^{rd}$ group

**(50) Answer :** (2)
**Solution:**
Electron gain enthalpy of He is lesser than Ar.

**(51) Answer :** (3)
**Hint:**
As the Bond order increases, the stability also increases.
**Solution:**

- $Li_2 \rightarrow Li_2^+$

$BO$ 1 0.5

$(Diamagnetic \quad (Paramagnetic)$

- $O_2 \rightarrow O_2^+$

$BO$ 2 2.5

$Paramagnetic \quad Paramagnetic$

- $N_2^+ \rightarrow N_2$

$BO$ 2.5 3

$paramagnetic \quad Diamagnetic$

- $H_2 \rightarrow H_2^+$

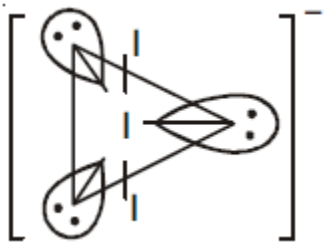
$BO$ 1 0.5

$Diamagnetic \quad Paramagnetic$

**(52) Answer :** (3)
**Solution:**

Hybridisation of central I atom is $sp^3d$ and three lone pairs are present at central atom.

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**(53)** **Answer :** (1)
**Solution:**
Standard molar enthalpy of formation of $Br_2(l)$ is zero at 298 K

**(54)** **Answer :** (2)
**Hint:**
$\Delta H = \Sigma B.E_{(reactant)} - \Sigma B.E_{(product)}$
**Solution:**
$\frac{1}{2}H_2 + \frac{1}{2}I_2 \rightarrow HI$
$\Delta H = \frac{1}{2}B.E._{(H_2)} + \frac{1}{2}B.E._{(I_2)} - B.E._{(HI)}$
or $BE_{(HI)} = \frac{1}{2}\left(B.E._{(H_2)} + B.E._{(I_2)}\right) - \Delta H$
$= \frac{1}{2}(435 + 150) - 26.5$ = 266 kJ/mole

**(55)** **Answer :** (4)
**Hint:**
$\Delta G = \Delta H - T\,\Delta\, S$
**Solution:**
$\Delta H = \Delta U + \Delta n_g RT$
$= 1.2 \times 10^3 + 2 \times 2 \times 400$
$= 1.2 \times 10^3 + 1.6 \times 10^3 = 2.8 \times 10^3$ cal
$\Delta G = \Delta H - T\Delta S$
$= 2.8 \times 10^3 - 400 \times 40 = -13.2$ kcal

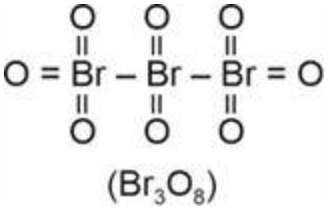
**(56)** **Answer :** (2)
**Hint:**
For any gaseous equilibrium
$K_P = K_C(RT)^{\Delta ng}$
**Solution:**
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ ; $K_P = K_C$
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ ; $K_P = K_C(RT)^{-2}$
$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ ; $K_P = K_C(RT)$
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ ; $K_P = K_C(RT)^{-1}$

**(57)** **Answer :** (2)
**Hint:**
pOH = 14 – pH
**Solution:**
Given pH=12
(or) $[H^+] = 10^{-12}$
$\therefore \left[OH^-\right] = \frac{10^{-14}}{10^{-12}} = 10^{-2}$
$Ba(OH)_2 \rightleftharpoons \underset{S}{\overset{2+}{Ba}} + \underset{2S}{2OH^-}$
$\left[OH^-\right] = 10^{-2}$
$2S = 10^{-2}$
$S = \frac{10^{-2}}{2}$
$= 4 \times \left(\frac{10^{-2}}{2}\right)^3$
$= 5 \times 10^{-7}$

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**(58)** **Answer :** (4)
**Solution:**
Addition of inert gas at constant volume does not affect the equilibrium.

**(59)** **Answer :** (2)
**Hint:**



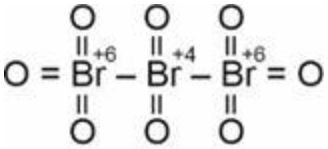
$(Br_3O_8)$

**Solution:**

- $\overset{-4+1}{C}H_4(g) + 2\overset{0}{O_2} \xrightarrow{\Delta} \overset{+4-2}{C}O_2(g) + 2\overset{+1}{H_2}O^{-2}(l)$

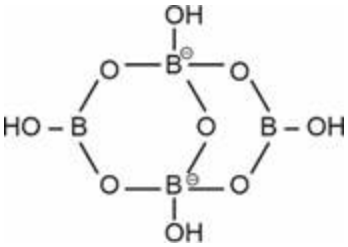
is an example of combination redox reaction

-



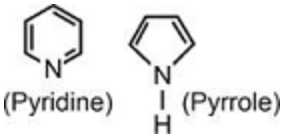
$(Br_3O_8)$

**(60)** **Answer :** (3)
**Hint:**
Borax contains 5 (B – O – B) bonds.
**Solution:**
The correct formula is $[B_4O_5(OH)_4]^{2-}$



**(61)** **Answer :** (2)
**Solution:**
Graphite is thermodynamically most stable allotrope of carbon.

**(62)** **Answer :** (3)
**Hint:**
Kjeldahl method is not applicable to compound containing nitrogen in nitro, azo groups and nitrogen present in the ring
**Solution:**



(Pyridine) (Pyrrole)

Kjeldahl method is not applicable to pyridine and pyrrole compounds.

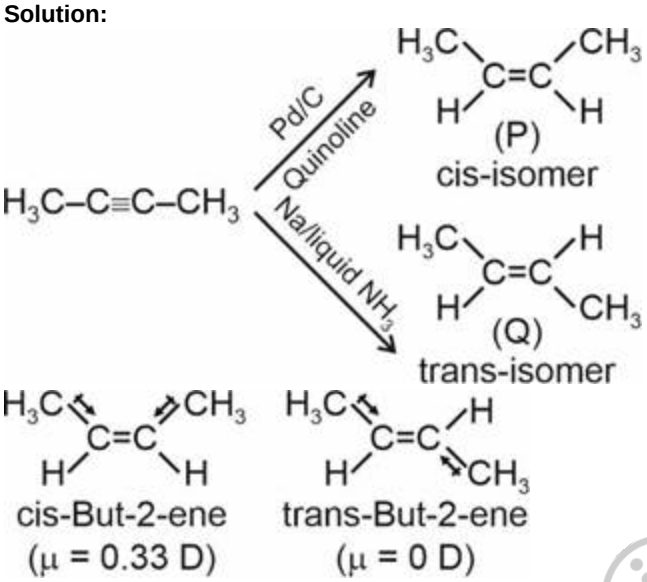
**(63)** **Answer :** (3)
**Hint:**
$NO_2$ group show –R effect as well as –I effect. Both effect decrease the stability of carbocation.
**Solution:**
Stability of carbocation
⇒ (IV) > (III) > (I) > (II)

**(64)** **Answer :** (1)
**Hint:**
More symmetrical is the compound, better is the crystal packing and higher is the melting point.
**Solution:**

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| Compound | Melting point (K) |
|---|---|
| Heptane | 182.4 |
| Hexane | 178.5 |
| 2, 2- Dimethylpropane | 256.4 |
| 2-Methylbutane | 113.1 |

**(65)** **Answer :** (3)
**Hint:**
Boiling point of cis form of alkene is more than the trans form of alkene.
**Solution:**

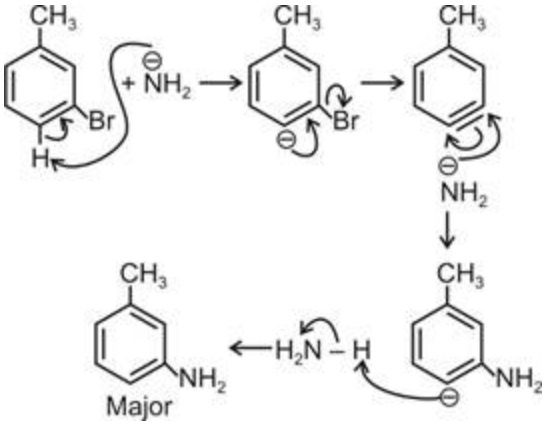



**(66)** **Answer :** (3)
**Hint:**
The compounds/species having (4n + 2)π electrons will obey Huckel's rule
**Solution:**


and

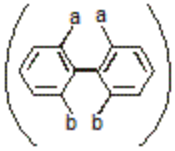
follow Huckel's rule.

**(67)** **Answer :** (2)
**Hint:**
Reaction follows benzyne mechanism.
**Solution:**




**(68)** **Answer :** (4)
**Hint:**
Biphenyls of type

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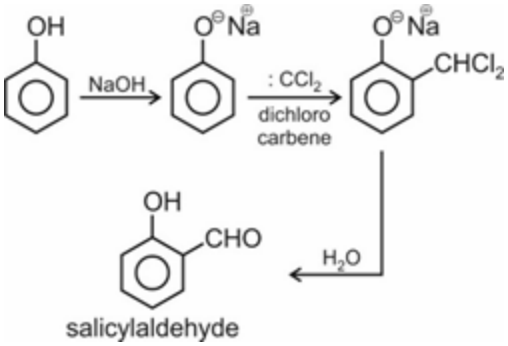

are optically active when a and b are bulky groups

**(69)** **Answer :** (4)

**Hint:**

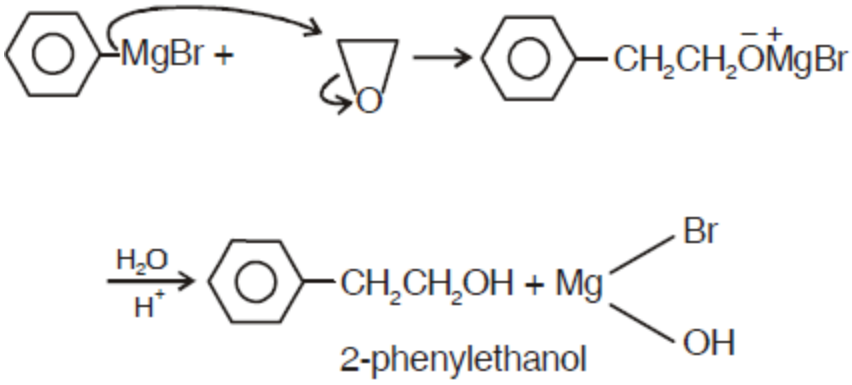
Reimer-Tiemann reaction.

**Solution:**




**(70)** **Answer :** (4)

**Hint:**

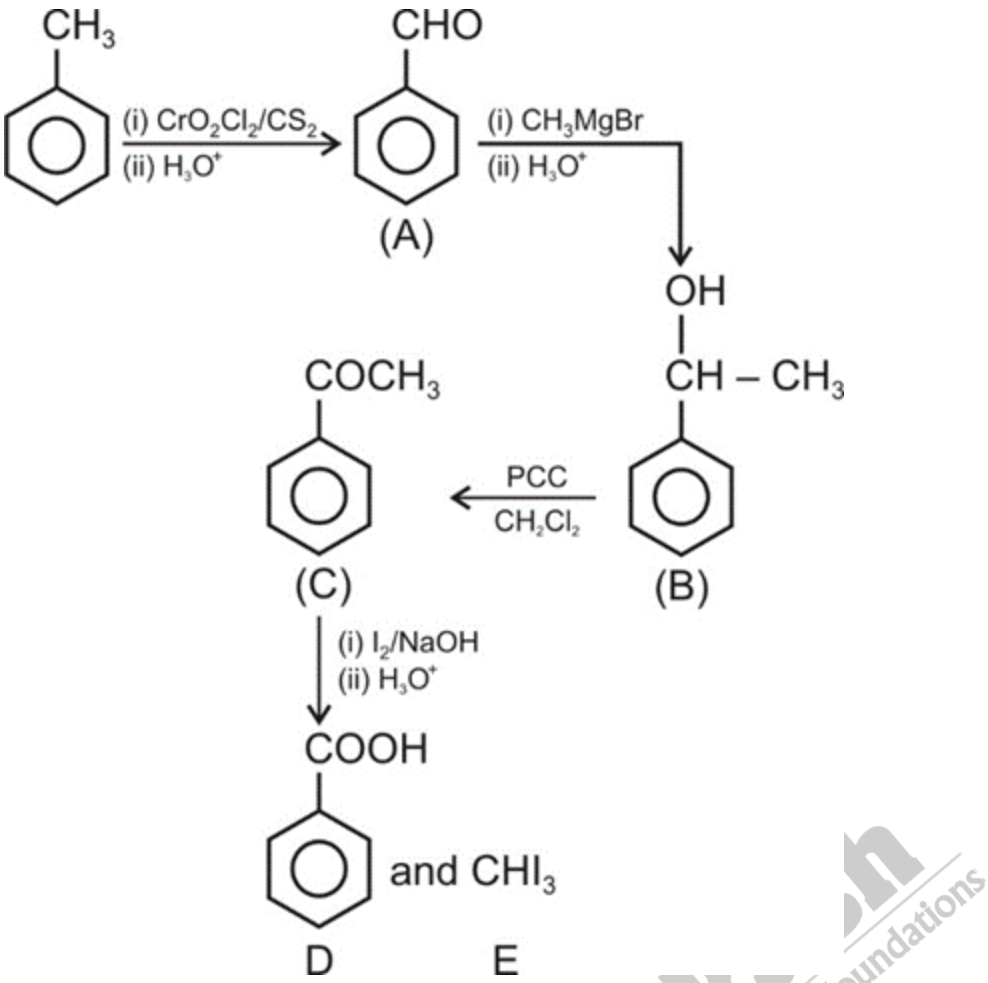



**(71)** **Answer :** (4)

**Hint:**

Reaction of Toluene with $CrO_2Cl_2/CS_2$ followed by $H_3O^+$is Etard oxidation.

**Solution:**

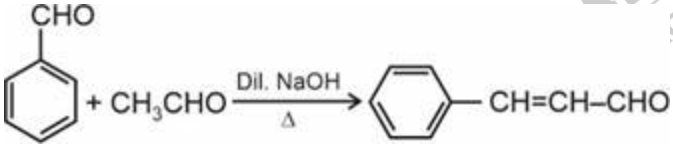



**(72)** **Answer :** (1)

**Hint:**

Ph–CH=CH–CHO is cinnamaldehyde

**Solution:**

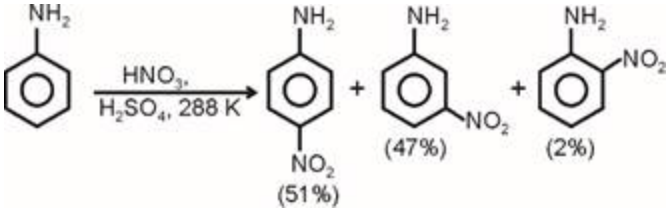



**(73)** **Answer :** (3)

**Hint:**

In the strongly acidic medium, aniline is protonated to form the anilinium ion, which is meta directing.

**Solution:**



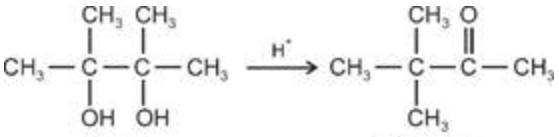

**(74)** **Answer :** (3)

**Hint:**

Ester having α-hydrogen on treatment with a strong base undergo self condensation to produce β-ketoesters.

**Solution:**

$2CH_3COOC_2H_5 \xrightarrow[(ii)\ H^+, H_2O]{(i)\ C_2H_5O^-Na^+} CH_3COCH_2COOC_2H_5$ is Claisen Ester condensation




is Pinacol-Pinacolone rearrangement

$CH_3-CH_2-COOH \xrightarrow[(ii)\ H_2O]{(i)\ Red\ P/Cl_2} CH_3-\underset{\underset{Cl}{|}}{CH}-COOH$ is Hell-Volhard Zelinsky reaction.

**(75)** **Answer :** (4)
**Solution:**
Highest $pK_b$ means weakest base.

**(76)** **Answer :** (4)
**Hint:**
Amylose constitutes about 15-20% of Starch.
**Solution:**
Amylopectin constitutes about 80-85% of starch.

**(77)** **Answer :** (1)
**Hint:**
Essential amino acid cannot be synthesised in body.
**Solution:**
Lysine is an essential amino acid.

**(78)** **Answer :** (4)
**Hint:**
A mixture of chloroform and acetone shows negative deviation from Raoult's law.
**Solution:**
• Acetone + carbon disulphide forms a positive deviation → non-ideal solution
• Benzene + Toluene → ideal solution
• n-hexane + n-heptane → ideal solution

**(79)** **Answer :** (2)
**Solution:**
$\Delta T_b \propto i \times c$
For glucose $1 \times 0.05 = 0.05$
For KCl $2 \times 0.02 = 0.04$
For $CaCl_2$ $3 \times 0.01 = 0.03$
For NaCl $2 \times 0.01 = 0.02$

**(80)** **Answer :** (3)
**Solution:**
Greater is the standard reduction potential, lesser is the reducing power.
Na
Mg
Al
Zn
$E^\circ_{Zn^{2+}/Zn} = -0.76\ V$ $E^\circ_{Al^{3+}/Al} = -1.66\ V$
$E^\circ_{Mg^{2+}/Mg} = -2.36\ V$ $E^\circ_{Na^{+}/Na} = -2.71\ V$
Order of reducing power
Na > Mg > Al > Zn

**(81)** **Answer :** (1)
**Hint:**
$2H^+(aq) + 2e^- \rightarrow H_2(g)$
$E_{H^+/H_2} = E^\circ_{H^+/H_2} - \frac{0.0591}{2}\log\frac{pH_2}{[H^+]^2}$
**Solution:**
$E_{H^+/H_2} = 0 - \frac{0.0591}{2}\log\frac{1}{(10^{-5})^2}$
$= -\frac{0.0591}{2}\log 10^{10}$
$= -\frac{0.0591}{2} \times 10 = -0.295\ V$

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**(82) Answer :** (3)

**Hint:**

Rate of reaction $\propto [A_3]^x [A_2]^y$

**Solution:**

$A_3 \rightleftharpoons A_2 + A,\ K_{eq} = \frac{[A_2][A]}{[A_3]} \Rightarrow [A] = K_{eq}\frac{[A_3]}{[A_2]}$

$A_3 + A \rightarrow 2A_2$,

rate = $K[A_3][A]$

rate = $K \times K_{eq}\frac{[A_3]}{[A_2]}[A_3]$

Rate = $K'[A_3]^2 [A_2]^{-1}$

$\therefore$ Overall order = (2) + (–1) = 1

**(83) Answer :** (3)

**Hint:**

Limiting molar conductivity of KCl is greater than NaCl.

**Solution:**

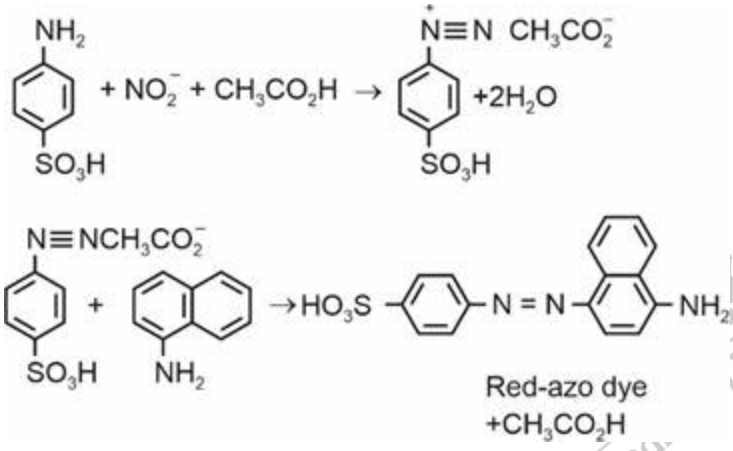
The value of $\wedge_m$ increases steeply for P on dilution due to increase in α and consequently the number of ions in total volume of solution.

**(84) Answer :** (4)

**Hint:**

This test is based upon the diazo-coupling reaction of sulphanillic acid. This test is specific for nitrite radical.

**Solution:**




**(85) Answer :** (3)

**Hint:**

For first order reaction,

$k = \frac{2.303}{(t_2 - t_1)} \log \frac{r_1}{r_2}$

$\frac{0.693}{t_{1/2}} = \frac{2.303}{(40-20)} \log \frac{0.04}{0.02} \Rightarrow t_{1/2} = 20$ sec

**Solution:**

For first order reaction,

$k = \frac{2.303}{(t_2 - t_1)} \log \frac{r_1}{r_2}$

$\frac{0.693}{t_{1/2}} = \frac{2.303}{(40-20)} \log \frac{0.04}{0.02} \Rightarrow t_{1/2} = 20$ sec

**(86) Answer :** (4)

**Hint:**

$Ce^{4+}$, $Lu^{3+}$ and $Yb^{2+}$ are diamagnetic.

**Solution:**

$Eu^{2+}$ : [Xe]$4f^7$
$Ce^{4+}$ : [Xe] $4f^0$
$Lu^{3+}$ : [Xe] $4f^{14}$
$Yb^{2+}$ : [Xe] $4f^{14}$

**(87)** **Answer :** (3)
**Solution:**
$Co^{3+}$ (high spin)
$d^6$ configuration = $t_{2g}^4 e_g^2$ (unsymmetrical)
$Fe^{3+}$ (low spin)
$d^5$ configuration = $t_{2g}^5 e_g^0$ (unsymmetrical)
so these complexes show Jahn Teller distortion.

**(88)** **Answer :** (4)
**Hint:**
cis-$[CrCl_2(ox)_2]^{3-}$ is optically active molecule.
**Solution:**
cis-$[CrCl_2(ox)_2]^{3-}$ lacks plane of symmetry and centre of symmetry, so, it is a chiral molecule.

**(89)** **Answer :** (3)
**Hint:**
Wilkinson catalyst $[(PPh_3)_3RhCl]$ is used for the hydrogenation of alkenes.
**Solution:**
• Chlorophyll is a coordination compound of magnesium
• Vitamin $B_{12}$, cyanocobalamin is a coordination compound of cobalt.
• Haemoglobin, the red pigment of blood which acts as oxygen carrier is a coordination compound of iron.

**(90)** **Answer :** (4)
**Hint:**
When $CuSO_4$ reacts with $K_4[Fe(CN)_6]$ a chocolate brown precipitate of copper (II) ferrocyanide is formed.
**Solution:**
$2CuSO_4 + K_4[Fe(CN)_6] \rightarrow Cu_2[Fe(CN)_6] + 2K_2SO_4$.

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## BIOLOGY

**(91)** **Answer :** (1)
**Solution:**
Cytoplasm is the main arena of cellular activities. It is the semi-fluid matrix and contains many cell organelles.

**(92)** **Answer :** (3)
**Solution:**
Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.

**(93)** **Answer :** (4)
**Solution:**
Centriole gives cartwheel appearance.

**(94)** **Answer :** (1)
**Solution:**
In the given diagram of plasma membrane, 'A' represents hydrophilic polar head.

**(95)** **Answer :** (2)
**Solution:**
Recombination occurs between non-sister chromatids of homologous chromosome. Bivalents are clearly seen during pachytene.

**(96)** **Answer :** (4)
**Solution:**
Bivalent-Clearly visible at pachytene. Chiasmata-becomes visible during diplotene. Syncytium is a multinucleated condition in coconut.

**(97) Answer :** (2)
**Solution:**
S phase is also known as synthesis phase, and it occurs between $G_1$ and $G_2$ phase.

**(98) Answer :** (2)
**Solution:**
Classification of organisms into various categories help in identifying similarities and dissimilarities among different kinds of organism.

**(99) Answer :** (2)
**Solution:**
Cyanobacteria lack flagellum throughout their life cycle.

**(100) Answer :** (2)
**Solution:**
*Neurospora* is extensively used in biochemical and genetic work.

**(101) Answer :** (1)
**Solution:**
Deuteromycetes are also called fungi imperfecti.

**(102) Answer :** (4)
**Solution:**
In silk cotton, palmately compound leaves are seen, where the tip of the petiole bears all the leaflets in the form of bunch or a cluster.

**(103) Answer :** (1)
**Solution:**
Drupe is a fleshy fruit which is formed by a monocarpellary ovary.

**(104) Answer :** (2)
**Hint:**
In china rose, alternate phyllotaxy, superior ovary and valvate aestivation is present in calyx.
**Solution:**
In china rose, monoadelphous androecium and axile placentation is present.

**(105) Answer :** (3)
**Solution:**
In gymnosperms the male and the female gametophytes do not have an independent existence as they remain within the sporangia retained on the sporophyte.

**(106) Answer :** (3)
**Solution:**
Alveoli of lungs are lined by simple squamous epithelium.

**(107) Answer :** (1)
**Solution:**
'**X**' is a ligament and it is a type of dense regular connective tissue in which the collagen fibres are present in rows between many parallel bundle of fibres. '**Y**' is adipose tissue and it is a type of loose connective tissue that is located mainly beneath the skin. Cells of this tissue are specialised to store fats. The excess of nutrients which are not used immediately are converted into fats and stored in this tissue.

**(108) Answer :** (4)
**Solution:**
Heart of cockroach consists of elongated muscular tube lying along mid dorsal line of thorax and abdomen.

**(109) Answer :** (3)
**Solution:**
Pancreas, a digestive gland produces pancreatic juice containing digestive enzymes. The duodenum receives pancreatic juice from pancreas through a common bile duct. Pancreatic juice digests proteins and carbohydrates.

**(110) Answer :** (4)
**Solution:**
Animals possessing coelom are called coelomates, *e.g.* annelids, molluscs, arthropods, echinoderms, chordates, *etc*.

**(111) Answer :** (2)
**Solution:**
Homocercal tail fin, swim bladder and operculum are found in bony fishes. Placoid scales and claspers are features of cartilaginous fishes.

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**(112) Answer :** (4)

**Solution:**

Hint : Chordates have dorsal nerve cord.

Sol. : True segmentation is also known as metamerism which is present in annelids, arthropods and chordates. Notochord is always formed dorsally in the body of chordates during embryonic development.

**(113) Answer :** (1)

**Hint:**

Tripeptide consists of three amino acids.

**Solution:**

First amino acid of chain is serine which is towards N-terminal.

Cytosine is the nitrogenous base.

Given structure is in primary configuration.

**(114) Answer :** (2)

**Solution:**

Oil, fats and cholesterol are included in lipids.

**(115) Answer :** (3)

**Solution:**

Ribs and sternum return to their original position during normal expiration.

**(116) Answer :** (2)

**Hint:**

In gymnosperm, the male and female gametophytes do not have an independent free-living existence.

**Solution:**

In gymnosperm, the multicellular female gametophyte is also retained within megasporangium.

The pollen grain is released from the microsporangium.

They are carried by air currents and come in contact with the opening of the ovules borne on megasporophylls.

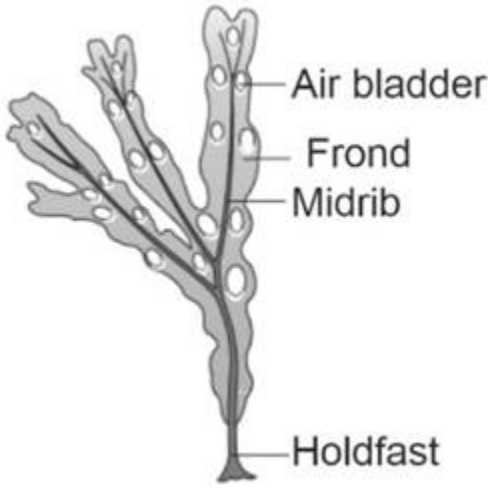
The pollen tube carrying the male gametes grows towards archegonia in the ovules and discharge their contents near the mouth of the archegonia.

Following fertilisation, zygote develops into an embryo and the ovules into seeds.

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**(117) Answer :** (2)

**Solution:**




**(118) Answer :** (2)

**Solution:**

Translocation of food is performed by vascular bundle of plants.

**(119) Answer :** (3)

**Solution:**

Vascular bundles lack cambium in leaves

**(120) Answer :** (4)

**Hint:**

Plastocyanin is placed after cytochrome $b_6f$ in Z-scheme.

**Solution:**

During chemiosmosis, plastocyanin transfer electrons from cyt $b_6f$ to PS I.

**(121) Answer :** (2)

**Solution:**
Ps-II is found in grana lamellae only.

**(122) Answer :** (2)
**Solution:**
Glycerol enters respiratory pathway after getting converted into glyceraldehyde 3 phosphate.

**(123) Answer :** (4)
**Solution:**
Cytochrome *c* is a small protein, which acts as a mobile carrier for transfer of electrons.

**(124) Answer :** (3)
**Solution:**
Auxin hormone prevent fruit and leaf drop at early stages but promotes abscission of older and mature leaves.

**(125) Answer :** (4)
**Solution:**
Ethylene promotes horizontal growth of the seedling.
Auxin promotes xylem differentiation.
Cytokinin promotes synthesis of chloroplast. Gibberellins induce bolting in plants.

**(126) Answer :** (2)
**Solution:**
Some members of Leguminosae, Rosaceae and Solanaceae have pollen viability period of several months.

**(127) Answer :** (4)
**Solution:**
Parthenocarpic fruits develop without the process of fertilization.
Dehydration and dormancy, help in storage of seeds.

**(128) Answer :** (2)
**Solution:**
Single cotyledon of monocot embryo is called scutellum.
Coconut water has free nuclei, (free nuclear endosperm)
Endosperm = 3n, perisperm = 2n
Hollow foliar structure enclosing leaf primordia is coleoptile.
Epiblast - remains of second cotyledon in monocots.

**(129) Answer :** (2)
**Solution:**
Generation of non-parental gene combination is called recombination while the physical association of gene on the same chromosomes is known as linkage.

**(130) Answer :** (1)
**Solution:**
50% of the male children will suffer with colorblindness and 50% with phenylketonuria.

**(131) Answer :** (2)
**Solution:**
$pO_2$ falls progressively in atmosphere with rise in altitude. This lowers the alveolar $pO_2$ and consequently reduces diffusion of $O_2$ from alveolar air to the blood.

**(132) Answer :** (3)
**Solution:**
**Hint:** Opening of the larynx is called glottis.
**Sol.:** During deglutition food can enter from pharynx into the larynx. This is prevented by closure of glottis with epiglottis.

**(133) Answer :** (3)
**Hint:**
Part of hindbrain
**Solution:**
The cardio-regulatory center of the brain is located in the medulla oblongata. The autonomous nerve fibres connected to the medulla oblongata work to control heart rate with the release of neurotransmitters. Neural signals *via* the sympathetic nerve fibres work to increase the heart rate and neural signals *via* parasympathetic nerve fibres work to decrease the heart rate.

**(134) Answer :** (2)
**Solution:**
Heart failure means the state of heart when it is not pumping blood effectively enough to meet the needs of the body.

**(135) Answer :** (4)

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**Solution:**
Secretion of inflammatory mediators is the function of basophils.

**(136) Answer :** (4)
**Solution:**
Crustaceans like Prawns have antennal glands or green glands for excretion. Protonephridia or flame cells are the excretory structures in platyhelminths, rotifers, some annelids and the cephalochordates. Nephridia are the tubular excretory structures of earthworms and other annelids. Malpighian tubules are the excretory structures of insects.

**(137) Answer :** (3)
**Solution:**
Reabsorption is minimum in ascending limb of Henle's loop.

**(138) Answer :** (1)
**Solution:**
Fibrous joints do not allow any movement. This type of joint is present between the flat skull bones. Pivot joint is present between atlas and axis while gliding joint is present between the carpals and saddle joint is present between carpal and metacarpal of thumb.

**(139) Answer :** (1)
**Solution:**
Each myosin filament is formed of many meromyosins. Each meromyosin has two parts namely, heavy meromyosin (HMM) and light meromyosin (LMM).

**(140) Answer :** (1)
**Hint:**
Part of brain containing iter.
**Solution:**
Hypothalamus is a part of forebrain.

**(141) Answer :** (1)
**Solution:**
Presence of myelin sheath on nerve fibres (axon) gives a white appearance.
The neuronal cell bodies are concentrated in the cerebral cortex and hence it appears grey in colour. The myelinated axons occur in the inner part of cerebrum which is called white matter.

**(142) Answer :** (2)
**Hint:**
It is produced from tyrosine and iodine.
**Solution:**
Thyroxine is derivative of amino acid tyrosine and is bound covalently to iodine.

**(143) Answer :** (4)
**Hint:**
Select a peptide hormone
**Solution:**
Peptide, polypeptide, protein hormones and catecholamines interact with the membrane bound receptors. They normally do not enter the target cell but generate second messengers triggering a cascade effect that amplifies the action of hormones significantly. Glucocorticoids, testosterone and estrogen are lipid soluble hormones that can cross the lipid bilayer of the plasma membrane and interact with intracellular receptors.

**(144) Answer :** (2)
**Solution:**
The penis is the male external genitalia. It is made up of special tissue that helps in the erection of the penis to facilitate insemination.

**(145) Answer :** (2)
**Solution:**
Oogonia and primary spermatocytes are diploid cells.

**(146) Answer :** (1)
**Solution:**
Near the nipple, mammary ducts expand to form mammary ampullae where some milk may be stored before going to lactiferous duct.

**(147) Answer :** (2)
**Solution:**
Answer (2)
Spermatocytogenesis the formation of spermatocyte which will be formed, hence will not be bypassed. Syngamy is fusion of gametes.

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**(148) Answer :** (2)
**Solution:**
ZIFT(zygote intra fallopian transfer) involves transfer of zygote or early embryos (with upto 8 blastomeres) into the fallopian tube.

**(149) Answer :** (3)
**Solution:**
The single step large mutation which can cause speciation was named saltation by de Vries.

**(150) Answer :** (4)
**Solution:**
In 1953, S.L. Miller, an American scientist created similar conditions in a laboratory scale (Figure 6.1). He created electric discharge in a closed flask containing CH4 , H2 , NH3 and water vapour at 800C. He observed formation of amino acids.

**(151) Answer :** (2)
**Solution:**
Bacteria like *Streptococcus pneumoniae* and *Haemophilus influenzae* are responsible for the disease pneumonia in humans which infects the alveoli (air filled sacs) of the lungs.

**(152) Answer :** (4)
**Solution:**
Rhino viruses represent one such group of viruses which cause one of the most infectious human ailments – the common cold. *Plasmodium*, a tiny protozoan is responsible malaria. *Entamoeba histolytica* is a protozoan parasite in the large intestine of human which causes amoebiasis.

**(153) Answer :** (3)
**Solution:**
The word AIDS stands for Acquired Immuno Deficiency Syndrome. There is lymphoid tissue also located within the lining of the major tracts (respiratory, digestive and urogenital tracts) called mucosa associated lymphoid tissue (MALT).

**(154) Answer :** (3)
**Hint:**
Same restriction enzyme produces same type of sticky ends.
**Solution:**
Restriction enzymes cut the strands of DNA a little away from the center of the palindromic sites, but between the same two bases on the opposite strands.
When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of 'sticky ends'.

**(155) Answer :** (1)
**Solution:**
Micro-injection is a vector-less method of gene transfer. In this method, the recombinant DNA is directly injected into the nucleus of an animal cell. In biolistics or gene gun (suitable for plants), cells are bombarded with high velocity micro-particles of gold or tungsten coated with DNA.

**(156) Answer :** (4)
**Solution:**
The morphological feature of *Drosophila* that makes it suitable for genetic studies is that the male and female flies are easily distinguishable. This allows researchers to easily identify and separate the sexes for controlled breeding experiments, which is crucial for genetic studies.

**(157) Answer :** (1)
**Hint:**
Unmodified allele represents the original phenotype, *i.e.,* dominant allele or wild type.
**Solution:**
The modified allele is generally the recessive allele or mutant type.

**(158) Answer :** (3)
**Solution:**
Free 2′OH of RNA makes it more labile and easily degradable.

**(159) Answer :** (4)
**Hint:**
3' end of tRNA attaches with an amino acid.
**Solution:**
This tRNA has UAC in anticodon loop, which means it will bind to the codon AUG of mRNA.

**(160) Answer :** (3)
**Solution:**
Helicase unwinds DNA helix.

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**(161) Answer :** (1)
**Solution:**
Length of DNA = total no. of bp × distance between two bp.

**(162) Answer :** (1)
**Solution:**
*Trichoderma polysporum* produces cyclosporin A (act as immunosuppressive agent).

**(163) Answer :** (2)
**Solution:**
BOD of a water body, increases when there is high amount of organic matter present in it and requires more aerobic bacteria to degrade the organic matter.

**(164) Answer :** (2)
**Solution:**
Visiting flamingoes and resident fishes compete for the common food, the zooplankton in the lake.

**(165) Answer :** (1)
**Solution:**
$N_{t+1} = N_t + [(\text{Natality} + \text{Immigration}) – (\text{Mortality} + \text{Emigration})]$

**(166) Answer :** (3)
**Solution:**
The effect of reduction in biodiversity was explained by Paul Ehrlich through Rivet popper hypothesis.
Term Biodiversity was popularized by Edward Wilson.

**(167) Answer :** (4)
**Solution:**
According to Rivet popper hypothesis,
Rivets refer to Species
Rivets on the wings are considered as Keystone species.

**(168) Answer :** (4)
**Solution:**

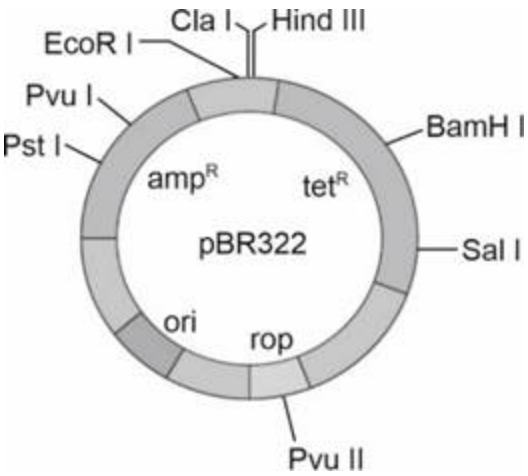
| Taxa | Number of species |
|---|---|
| Birds | 1300 |
| Reptiles | 378 |
| Fishes | 3000 |
| Mammals | 427 |

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**(169) Answer :** (3)
**Solution:**
Net primary productivity (NPP) is the available biomass for the consumption to heterotrophs *i.e.* herbivores and decomposers. It depends upon various environmental factors. NPP of terrestrial ecosystems is generally higher than that of ocean.

**(170) Answer :** (4)
**Solution:**
These food chains exemplify grazing food chain, which forms the major conduit of energy flow in aquatic ecosystem.
Secondary production of frog in food chain I is same as the frog in food chain II.

**(171) Answer :** (1)
**Solution:**




**(172) Answer :** (1)

**Solution:**

Enzymes are divided into 6 classes each with 4-13 subclasses and named accordingly by a four-digit number
Class I – Dehydrogenases
Class II – Transferases
Class III – Hydrolases
Class IV – Lyases
Class V – Isomerases
Class VI – Ligases

**(173) Answer :** (3)

**Hint:**

α-lactalbumin is a human milk protein.

**Solution:**

Bt cotton – High yield and pest resistant.
Golden rice – Enriched with vitamin A.
Rosie – Transgenic cow producing human protein (α-lactalbumin) enriched milk.
Flavr savr –Tomatoes having increased shelf life.

**(174) Answer :** (1)

**Solution:**

C-peptide chain is not present in mature insulin and disulphide bonds are formed during maturation into active insulin.

**(175) Answer :** (3)

**Hint:**

Sperm count × 60% × 40%

**Solution:**

Sperm count = 300 million / ejaculation
Because sperms with normal shape and size are 60% of sperm count and 40% of them show vigorous motility.
So sperm with normal shape, size and vigorous motility.
= $\left[300 \times \frac{60}{100} \times \frac{40}{100}\right]$ million / ejaculation
= 72 million / ejaculation

**(176) Answer :** (4)

**Solution:**

Seminal vesicles and bulbourethral glands are paired male accessory glands. Prostate is an unpaired male accessory gland.

**(177) Answer :** (2)

**Solution:**

The secretions of bulbourethral glands also helps in the lubrication of the penis. Leydig cells synthesise and secrete testicular hormones called androgens.

**(178) Answer :** (4)

**Solution:**

Answer (4)
– *Petromyzon* (Lamprey) and *Myxine (Hagfish)* are example of cyclostomes.

**(179) Answer :** (2)

**Solution:**

Lower mammals having snout are without heterodont dentition.

**(180) Answer :** (3)

**Hint:**

Proteins are polymers of different repeating units.

**Solution:**

Structural homopolymers include cellulose and chitin. Inulin and glycogen are storage homopolymers whereas collagen and RuBisCO are heteropolymers.

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